Shape Detective: Any Way It Turns
Answer Key
Grade 1

Quick Answers

1. (a) number of straight sides = 3, —
2. (a) number of sides = 4, —
3. (a) number of sides = 4, (b) number of corners = 4, —
4. (a) number of corners = 3, —
5. 2
6. (a) number of sides = 4, —
7. (a) number of corners = 3, —
8. (a) corners in all = 7, —

What is verified

1 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 1, 2, 3, 4, 6, 7, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 1
K.G – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–4 of 5 across 16 tagged checks.

1. A shape with three straight sides, sitting flat. (a) Sides? (b) Name?
Solution: (a) Trace each straight side with a finger and count: three sides. $\boxed{3}$
(b) *Instructor judges:* three straight sides makes it a **triangle**.

2. A tilted shape standing on a corner. (a) Sides? (b) Name?
Solution: (a) Count the sides: four, all the same length, with square corners. $\boxed{4}$
(b) *Instructor judges:* it is still a **square**. Tilting a shape does not change what it is — only how it sits on the page.

3. A tipped-over shape. (a) Sides? (b) Corners? (c) Name?
Solution: (a) Count the sides: $\boxed{4}$
(b) Each pair of sides meets at a corner — count them: $\boxed{4}$
(c) *Instructor judges:* opposite sides are equal and the corners are square, so it is a **rectangle**, even lying at a slant.

4. A shape pointing down. (a) Corners? (b) Name?
Solution: (a) Count where the sides meet: three corners. $\boxed{3}$
(b) *Instructor judges:* it is a **triangle** — pointing down does not change the name.

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5. Hexagon: 6 sides. Square: 4 sides. How many more sides has the hexagon?
Solution: Compare the counts: $6 - 4 = 2$. The hexagon has $\boxed{2}$ more sides.

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6. Ben says the tilted shape is a diamond, not a square. (a) Sides? (b) Is Ben right?
Solution: (a) Count the sides: $\boxed{4}$
(b) *Instructor judges:* Ben is not right. The shape has 4 equal sides and square corners, so it is a square — a tilted square is still a square. “Diamond” only describes how it is sitting.

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7. Kim’s shape has 3 straight sides, drawn long, skinny, and tilted. (a) Corners? (b) Name?
Solution: (a) A shape with 3 straight sides always has 3 corners — every pair of sides meets once. $\boxed{3}$
(b) *Instructor judges:* it is a **triangle**. Long, skinny, or tilted, the name comes from the 3 sides, not the look.

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8. Challenge: square (4 corners) and triangle (3 corners). (a) Corners all together? (b) Does turning change a name?
Solution: (a) Add the two counts: $4 + 3 = 7$. $\boxed{7}$
(b) *Instructor judges:* no — turning never changes a shape’s name. The name comes from the number of sides and corners, and turning changes neither.